

Week 1 Progress Report

Comparing LLM-Based Policy Generation and Reinforcement Learning for Embodied Control

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Week 1

1 Goal of Week 1

The goal of the first week was to set up a minimal working experimental pipeline for the project. I focused on one simple continuous-control task first, instead of immediately starting with the full LLM-based method. The main objective was to check whether I could create an environment, evaluate controllers, train a reinforcement learning baseline, and save results in a structured way.

2 Environment

The first environment used was **Pendulum-v1** from Gymnasium. I chose this environment because it has continuous observations and continuous actions, which makes it closer to the embodied-control setting of the project than a discrete task such as **CartPole**. At the same time, it is still simple enough to debug quickly. The observation contains:

- $\cos(\theta)$, the cosine of the pendulum angle;
- $\sin(\theta)$, the sine of the pendulum angle;
- $\dot{\theta}$, the angular velocity.

The action is one continuous torque value in the interval $[-2, 2]$.

3 Implemented Methods

Three methods were evaluated during Week 1.

3.1 Random Controller

The random controller samples a torque uniformly from the allowed action range at each time step. It does not use the observation and does not learn. This controller is used as a lower sanity-check baseline.

3.2 Manual Controller

The manual controller is a fixed hand-written rule. It uses the pendulum angle and angular velocity to compute a torque. The angle is reconstructed from the observation using

$$\theta = \text{atan2}(\sin(\theta), \cos(\theta)).$$

The controller then applies a simple proportional-derivative style rule:

$$u = -2.0\theta - 0.5\dot{\theta}.$$

Here, u is the torque applied to the pendulum. The value of u is clipped to the valid torque range $[-2, 2]$. This controller is not trained. It is only a basic manually designed policy.

3.3 PPO Baseline

I also trained a first Proximal Policy Optimization baseline using Stable-Baselines3. This was included to verify that the project can compare fixed controllers with a trained reinforcement learning policy. The PPO baseline was trained for 300,000 timesteps and then evaluated with 100 episodes, the same as the fixed controllers.

4 Evaluation Setup

All methods were evaluated on **Pendulum-v1**. Each reported result uses 100 evaluation episodes with seed 42.

The main metric is the mean cumulative reward over the evaluation episodes. In **Pendulum-v1**, rewards are usually negative, so values closer to zero are better.

5 Results

Table 1 shows the Week 1 results.

Table 1: Week 1 results on **Pendulum-v1**. Higher reward is better.

Method	Episodes	Mean reward	Std. reward	Training
Random controller	100	-1248.195	234.175	No
Manual controller	100	-1180.92	271.625	No
PPO baseline	100	-195.086	123.117	Yes

The manual controller performed better than the random controller. This means that the manual rule uses some useful information from the state. However, the improvement is still small compared to the PPO result. The PPO baseline performed much better than both fixed controllers. Its mean reward was approximately -195.086 , while the manual controller obtained -1180.920 and the random controller obtained -1248.195 .

6 Discussion

The Week 1 results show that the experimental pipeline works. The random controller performs poorly, as expected. The manual controller improves over random, which confirms that the evaluation setup can detect a difference between a purely random policy and a simple state-based controller.

The PPO result is much stronger. This is expected, because PPO actually learns from interaction with the environment, while the random and manual controllers are fixed. The result also confirms that the reinforcement learning training and evaluation code is working.

At this stage, the PPO result should still be seen as a preliminary baseline. Only one seed and one training budget were used. Later experiments should include multiple seeds and possibly longer training runs. Still, this result is useful because it gives a clear reference point for future LLM-generated controllers. A generated controller that performs close to the manual controller would be weak. A generated controller that moves toward the PPO result would be more interesting.

7 Conclusion

Week 1 established a working experimental setup for the project. I can now create a Gymnasium environment, evaluate fixed controllers, train a PPO baseline, save results, and compare policies using the same evaluation protocol.

The results show a clear ranking:

$$\text{PPO} > \text{Manual controller} > \text{Random controller}.$$

This gives a useful starting point for Week 2.